

Tutorial 4 – Create Tile Movement

First, create a script folder and create a new C# script with the name 'Tile Movement' as it'll manage the movement behaviour for each tile in the jigsaw puzzle. Then, declare a public property called 'Tile' as it represents the specific tile will control its movement.

The public class will handle the movement behaviour for each tile in our jigsaw puzzle called tile as it represents the specific tile that this script will control. Then define a private variable called mOffset.

```
Unity Script 0 references
public class TileMovement : MonoBehaviour
{
    2 references
    public Tile tile { get; private set; }
    private Vector3 mOffset = new Vector3(0.0f, 0.0f, 0.0f);
}
```

Afterwards, we will define a private method called GetCorrectPosition as it'll calculate and return the correct position for the tile based on its xindex and yindex.

```
private Vector3 GetCorrectPosition()
{
    return new Vector3(tile.xIndex * mOffset.x, tile.yIndex * 100f, 0f)
}
```

Next, we will implement the OnMouseDown as it is when the mouse button is pressed down while the cursor is over the GameObject to which this script is attached. In this method, we will set the offset between the mouse position and the tile's position.

```
private void OnMouseDown()
{
    if(EventSystem.current.IsPointerOverGameObject())
    {
        return;
    }

    mOffset = transform.position - Camera.main.ScreenToWorldPoint(new Vector3(Input.mousePosition.x, Input.mousePosition.y, 0.0f));
}
```

Next, we will implement the OnMouseDrag as the mouse will be dragged while the cursor is over the GameObject which is attached.

```
private void OnMouseDown()
{
    if (EventSystem.current.IsPointerOverGameObject())
    {
        return;
    }

    Vector3 curScreenPoint = new Vector3(Input.mousePosition.x, Input.mousePosition.y, 0.0f);
    Vector3 curPosition = Camera.main.ScreenToWorldPoint(curScreenPoint) + mOffset;
    transform.position = curPosition;
}
```

Afterwards, we will implement the OnMouseUp which the mouse button is released after being pressed down while the cursor is over the GameObject that this script is attached to. Inside this method.

```
private void OnMouseUp()
{
    if(EventSystem.current.IsPointerOverGameObject())
    {
        return;
    }

    float dist = (transform.position - GetCorrectPosition()).magnitude;
    if(dist < 50f)
    {
        transform.position = GetCorrectPosition();
    }
}
```

Now, we will set the script component to each tile as it'll be applied onto the BoardGen, go to the CreateGameObjectFromTile function and add the TileMovement component to the tile game object.

```
public static GameObject CreateGameObjectFromTile(Tile tile)
{
    GameObject obj = new GameObject();

    obj.name = "TileGameObje_" + tile.xIndex.ToString() + "_" + tile.yIndex.ToString();

    obj.transform.position = new Vector3(tile.xIndex * Tile.tileSize, tile.yIndex * Tile.tileSize, 0.0f);

    SpriteRenderer spriteRenderer = obj.AddComponent<SpriteRenderer>();
    spriteRenderer.sprite = SpriteUtils.CreateSpriteFreeTexture2D(
        tile.finalCut,
        0,
        0,
        Tile.padding * 2 + Tile.tileSize,
        Tile.padding * 2 + Tile.tileSize);

    BoxCollider2D box = obj.AddComponent<BoxCollider2D>();
    return obj;
}
```

Go to the Unity Editor and click play as it'll allow you to move the tiles around and drag them into different locations.

Final Script:

Tile Movement

```
using UnityEngine;
using UnityEngine.EventSystems;

@ Unity Script 0 references
public class TileMovement : MonoBehaviour
{
    2 references
    public Tile tile { get; private set; }
    private Vector3 mOffset = new Vector3(0.0f, 0.0f, 0.0f);

    2 references
    private Vector3 GetCorrectPosition()
    {
        return new Vector3(tile.xIndex * mOffset.x, tile.yIndex * 100f, 0f)
    }

    @ Unity Message 0 references
    private void OnMouseDown()
    {
        if(EventSystem.current.IsPointerOverGameObject())
        {
            return;
        }

        mOffset = transform.position - Camera.main.ScreenToWorldPoint(new Vector3(Input.mousePosition.x, Input.mousePosition.y, 0.0f));
    }

    @ Unity Message 0 references
    private void OnMouseDrag()
    {
        if (EventSystem.current.IsPointerOverGameObject())
        {
            return;
        }

        Vector3 curScreenPoint = new Vector3(Input.mousePosition.x, Input.mousePosition.y, 0.0f);
        Vector3 curPosition = Camera.main.ScreenToWorldPoint(curScreenPoint) + mOffset;
        transform.position = curPosition;
    }

    @ Unity Message 0 references
    private void OnMouseUp()
    {
        if(EventSystem.current.IsPointerOverGameObjects())
        {
            return;
        }

        float dist = (transform.position - GetCorrectPosition()).magnitude;
        if(dist < 50f)
        {
            transform.position = GetCorrectPosition();
        }
    }
}
```